

Site Name -

Pre Commissioning Checklist for KAVACH Building Works

1st Stage Inspection – At the time of PCC/ Post PCC

#	Check Points	Observations
1	Date of Inspection	
2	Size of building	
3	Structural Drawing Number	
4	Excavation Layout Line- Check whether excavation has been done as per approved structural drawing	
5	Depth of foundation	As per drawing- Actual-
6	PCC thickness	
7	Cement – Make, Grade, Month and Year of Manufacture (as per design make given by Agency)	
8	Check that PCC is done with at least M10 grade concreting (Nominal Mix- 50 Kg Cement, 160 Kg Fine Aggregate, 320 Kg Coarse	
9	Curing of PCC	
10	Back filling with layers for proper water compaction	

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2ND Stage Inspection – Reinforcement Checking

#	Check Points	Observations
1	Centre Line Alignment of Footings / Pedestals & columns (As per drawing)	
2	Reinforcement make	
3	Bar cutting and bending (BBS schedule) as per item like lintel, beam, roof slab, stair case	As per BBS Drawing
4	Site in Dry condition / Water Logged Area	
5	Reinforcement placement as per drawing in Main Footing	
6	Reinforcement placement as per drawing in Footing, column beam or slab	
7	Column & beam Formwork / shuttering plumb	
8	Cover block Adequacy	
9	Mode of Concreting planned (RMC/ Weight vatcher) as per site condition	
10	Availability of Material for casting (if not RMC). Material approved by officers as per site location	
11	Availability of water for casting (if not RMC). Railway water/private arrangement	
12	Any other comments -	
13	Date of Inspection	

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3rd Stage Inspection – Casting of Footing, column, beam, lintel, slab, stair case up to Roof level

#	Check Points	Observations
1	Cement – Make, Grade, Month and Year of Manufacture (as per design make given by Agency, as per site location wise)	
2	Cement – Quantity Adequate	
3	Coarse / Fine Aggregates – Quantity / Quality Adequate	
4	Steel Shuttering Quality and Alignment	
5	Mix Proportion adopted as per design mix	
6	Mode of Concreting (RMC/ Weight vetcher)	
7	Water cement Ratio & Vibrator (as per design mix allowed)	
8	Slump Test – At Start	
9	Slump Test – At mid point	
10	Footing, column, beam (as per clause 4.6.6)	
11	Centre Line of footing, column, beam in plumb check of Shuttering	
12	Curing of footing, column, beam, stair case, brick work, plaster, etc.	
13	Any other comments -	
14	Date of Inspection	

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4th Stage Inspection – RCC work & BBS as per Drawing

#	Check Points	Observations
1	Plinth beam level required as per drawing height	
2	Door opening of over plinth beam as per drawing size.	
3	Ventilator height from floor level as per drawing	
4	Plum check of column, beam shuttering – Pre-casting	
5	Cement – Make, Grade, Month and Year of Manufacture (as per design mix)	
6	Cement – Quantity Adequate	
7	Coarse / Fine Aggregates – Quantity / Quality Adequate	
8	Mx Proportion adopted (as per design mix)	
9	Mode of Concreting (RMC/ Weight vatcher)	
10	Water cement Ratio & Vibrator (as per design mix)	
11	Slump Test – At Start	
12	Slump Test – At mid point	
13	Plumb check of columns Post	
14	Cube filling (ensure 3 numbers) columns, beam, roof slab- (ref clause 4.6.6)	
15	Curing of footing, columns, beam, lintel, roof slab	
16	Door & window ventilator as per drawing & Railways specifications	
17	White washing/ colour washing/ distempering & painting as per Railway norms directed by Engineer Inchagre.	
18	Any other comments -	
19	Date of Inspection -	

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INSPECTION CHECK LIST FOR KAVACH BUILDING

The KAVACH building will be inspected at each stage by the inspections from WC Railway and firm as per the following inspection plan. Clearance will be given only after these tests are passed.

Sl. No.	Description of Items to be checked	OK / Not OK	Remarks
I	Excavation of Pit, Layout and Kilometre Chainage. a) Correct marking of pit location with reference to bench marks. b) Checking of alignment and measurement of the pit before PCC		
II	Quality of materials		
a)	CEMENT: standard make (OPC- 43 grade) ISI marked		
b)	FINE AGGREGATE (Sand): Be clean & shall not contain clay or any deleterious material		
c)	Coarse aggregate)Jelly/ Kankar of 20 mm) (a) flaky or elongated (b) nearly cubic shape		
d)	Steel Rods used for Reinforcement: As per IS 1786, Fe415/Fe/500D		
e)	Railway/ Private water used shall be clean and drinking quality and shall not be salty.		
III	RCC Stages		
a)	Formwork Checking (a) Formwork is properly aligned (b) Check if the formwork is water- tight, (c) Supports are properly placed and fixed, (d) Check that width, depth and height of form work are correct, as per design & drawing.		
b)	Reinforcement Checking (a) Steel rods are placed as per drawing, (b) Check reinforcement bars are straight as per BBS. (c) Dia of reinforcement bars is as per design as per BBS, (d) Spacing of reinforcement bar is as per design as per BBS. (e) Lap length is adequate, (f) Check the clear cover, (g) Cover blocks are enough, (h) Check quantity count BBS as per drawing.		
c)	Pre-concreting Checking (a) Check all required materials and tools are available at site (b) Check that adequate lighting is available (if it is planned to work at night) (c) Check the form work supports are intact and not displaced.		

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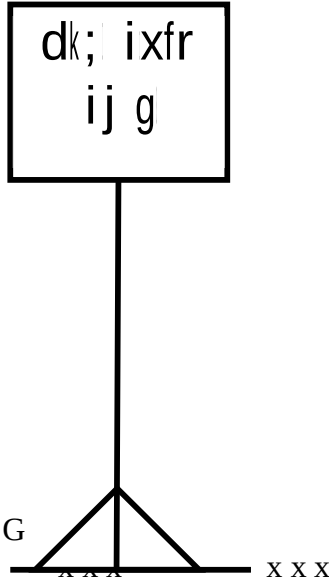
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d)	Checking during concreting (a) Check slump for every 20 cu.m of concrete for proper water and cement ratio. (b) Concrete cubes are made for testing (Concrete shall be collected from 3 or 4 random mixes for making concrete cubes (150 mm) which would be sent for third party examination or strength testing at site. (c) Check that vibrator is used adequately and properly (d) Check that concrete surface is level (e) Check the finishing of concrete surface is level (f) Check that adequate curing is done		
IV	Check for escarpment: There is no such local site condition which necessitate change in design of concrete surface of foundation		
V	All building & tower pits, all excavated site is barricaded		
VI	Display board for site working in progress- at each site 		
VII	Curing of concreting		
VIII	Back filling with layers for proper water compaction		
IX	Electrical works a) Check that all electrical fittings are concealed b) Check that all electrical fittings are as per drawing c) Check that appropriate segregation is available between signalling cables and electrical cables		

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X	Finishing works a) Check that tiling of floors and skirting has been done as per drawing b) Check that whitewashing/distempering has been done c) Check that all holes/doors/windows are sealed to prevent entry of rodents d) Check that there is no gap below doors of KAVACH rooms to prevent entry of rodents and dust e) Check that double lock arrangement has been provided on doors f) Check that drainage arrangement is proper g) Check that checkered tiles have been provided on stairs and verandah on the outside of KAVACH building h) Check that railings have been provided on the staircases		
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